

AI-Powered Real-Time Customer Support Coaching Assistant

Project Documentation

Prepared for the Partial Fulfilment of

Infosys Springboard Virtual Internship Program 7.0

by

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Batch No: 1

Project Code: #Z-1-1

Program Duration: 8 Weeks (4 Milestones)

July, 2026

Table of Contents

- 1. Project 3**
 - 1.1 Problem Statement 3**
 - 1.2 Project Objective 3**
 - 1.3 Interaction Modes 3**
 - 1.4 Key Outcomes 3**
 - 1.5 Key Features 4**
 - 1.6 User Roles 4**
- 2. Tools and Technology Stack 4**
- 3. System Architecture 5**
 - 3.1 Frontend Layer 5**
 - 3.2 Backend Layer 6**
 - 3.3 Multi-Agent AI Systems 6**
 - 3.4 Data Storage Layer 7**
 - 3.5 External AI Services 7**
 - 3.6 Deployment Architecture 7**
 - 3.7 End-to-End Workflow 7**
- 4. Schema and Data Models 8**
 - 4.1 MongoDB Atlas — Collections 8**
 - 4.2 ChromaDB — Vector Collection 9**
 - 4.3 Firebase Authentication 10**
- 5. Agent Specifications 10**
- 6. Functional Modules 11**
- 7. Project Phases 11**
- 8. Evaluation Criteria 12**
- 9. Glossary 13**

1. Project Overview

1.1 Problem Statement

Customer service representatives handle dozens of live support interactions every day, often without access to real-time guidance, relevant knowledge resources, or immediate feedback on the quality of their responses. Traditional training relies on post-call reviews, which are slow and reactive and fail to improve agent performance during live interactions, where it matters most.

1.2 Project Objective

The objective of this project is to build an AI Customer Support Coaching Assistant — a platform that coaches customer service agents in real time during live, text-based support interactions. A multi-agent pipeline analyses each conversational turn, retrieves relevant knowledge base content, suggests responses and tone improvements, and monitors escalation risk, all surfaced through a three-panel coaching console. After each interaction, the system produces a structured performance report.

1.3 Interaction Modes

- **Simulator Mode** — a Customer Simulator Agent generates realistic customer messages turn by turn, based on a configurable scenario and persona.
- **Manual Mode** — the agent pastes in incoming customer messages, and the coaching pipeline analyses them in real time.
- **Replay Mode** — a pre-loaded support transcript is replayed message by message for retrospective coaching.

1.4 Key Outcomes

- A multi-agent real-time coaching pipeline that analyses each conversation turn and delivers instant guidance.
- A Customer Simulator Agent that generates realistic, scenario-based customer messages.
- RAG-powered knowledge recommendations — relevant FAQs, support articles, and troubleshooting steps.

- **Continuous escalation risk detection with reasoning and recommended intervention strategies.**
- **Post-interaction performance reports with sentiment journey, quality scoring, and coaching recommendations.**

1.5 Key Features

- Real-time AI reply suggestions
- Intent detection and Sentiment analysis
- RAG-based knowledge retrieval
- Coaching and tone improvement
- Quality score and monitoring
- Conversation summarisation
- Escalation detection
- Analytics and reporting

1.6 User Roles

Role	Responsibility
Support Agent	Handles live customer conversations and receives real-time AI coaching.
Team Lead	Monitors team performance and reviews aggregated coaching outcomes.
Quality Analyst	Analyses conversations and quality scores for compliance and improvement.
Admin	Manages users, permissions, and overall system settings.

2. Tools and Technology Stack

Layer	Technology
Frontend	React.js, Vite, Tailwind CSS, React Router DOM, Lucide React (Icons), JavaScript (ES6+), Fetch API
Backend	FastAPI, WebSockets (Python), Flask, Flask-CORS, REST APIs (JSON)
AI Orchestration	LangChain
LLM Providers	OpenAI GPT-4.1 or Google Gemini API or Groq API (llama-3.3-70b-versatile)
Primary Database	MongoDB Atlas, PostgreSQL, SQLAlchemy
Vector Database	ChromaDB (1.5.9)

Embedding Model	Sentence Transformers, HuggingFace Embeddings, ChromaDB's built-in DefaultEmbeddingFunction
Authentication	Firebase Authentication, JWT Authentication, Role-Based Access Control
Deployment	Vercel, Docker, Docker Compose, Render, Vercel, AWS Ready

3. System Architecture

The platform is a layered, event-driven system. A React-based frontend communicates over WebSockets with a FastAPI backend, which orchestrates a four-stage multi-agent AI pipeline. Conversation, knowledge, and authentication data are persisted across three specialised data stores, and the system integrates external LLM and embedding services to support its AI capabilities.

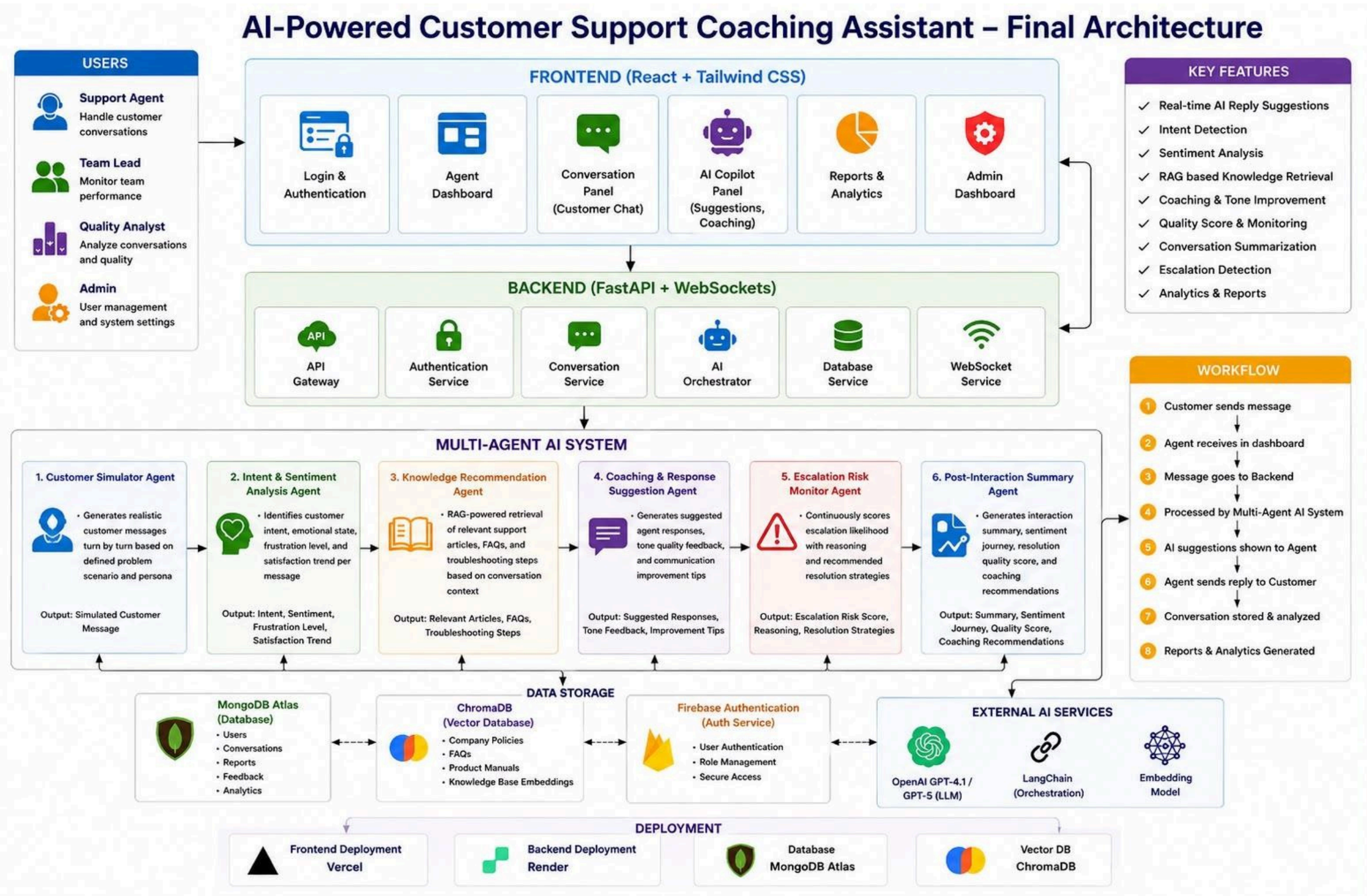


Figure 1: AI-Powered Customer Support Coaching Assistant: Final Architecture

3.1 Frontend Layer — React + Tailwind CSS

The frontend is a single-page application organised into six functional panels:

- **Login and Authentication** — secure sign-in and session management.
- **Agent Dashboard** — the support agent's home view and session entry point.
- **Conversation Panel (Customer Chat)** — the live or replayed conversation window.
- **AI Copilot Panel** — real-time suggestions and coaching guidance.

- **Reports and Analytics** — post-interaction reports and cross-session trends.
- **Admin Dashboard** — user management and system configuration.

3.2 Backend Layer — FastAPI + WebSockets

The backend exposes a FastAPI service layer and maintains persistent WebSocket connections to stream coaching output to the frontend in real time. It comprises six services:

Service	Function
API Gateway	Single entry point that routes incoming requests to the appropriate backend service.
Authentication Service	Handles login, session tokens, and role-based access control.
Conversation Service	Manages conversation state across Simulator, Manual, and Replay modes.
AI Orchestrator	Coordinates the four-agent pipeline and aggregates their outputs per turn.
Database Service	Handles reads and writes to MongoDB Atlas for persistent application data.
WebSocket Service	Streams live coaching updates, suggestions, and alerts to the frontend.

3.3 Multi-Agent AI System

A four-stage multi-agent pipeline processes every conversational turn. Each agent has a distinct responsibility and a well-defined output that feeds the next stage and the coaching console.

Agent	Core Functions	Output
Customer Understanding Agent	Intent Detection, Sentiment Analysis, Context Understanding	Intent, Sentiment, Context, Entities
Knowledge Agent (RAG Agent)	RAG Retrieval, Semantic Search, Knowledge Recommendation	Relevant Documents, Policies, FAQs

Response & Coaching Agent	AI Reply Suggestions, Tone Improvement, Coaching Tips	Suggested Reply, Improved Tone, Coaching Tips
Quality Monitoring Agent	Escalation Detection, Quality Score, Conversation Summarisation	Quality Score, Escalation Risk, Summary

3.4 Data Storage Layer

Store	Contents
MongoDB Atlas (Database)	Users, Conversations, Reports, Feedback, Analytics.
ChromaDB (Vector Database)	Company policies, FAQs, product manuals, and knowledge-based embeddings for RAG retrieval.
Firebase Authentication (Auth Service)	User authentication, role management, and secure access control.

3.5 External AI Services

- OpenAI GPT-4.1 / GPT-5 — the underlying large language models used for reasoning, generation, and analysis across all agents.
- LangChain — the orchestration framework coordinating prompts, tool calls, and agent hand-offs.
- Embedding Model — generates vector embeddings for knowledge-base content and semantic search queries against ChromaDB.

3.6 Deployment Architecture

Component	Deployment Target
Frontend	Vercel
Backend	Render
Database	MongoDB Atlas
Vector Database	ChromaDB

3.7 End-to-End Workflow

1. Customer sends a message.
2. The agent receives the message in the dashboard.
3. Message goes to the backend via the API Gateway / WebSocket Service.
4. The message is processed by the Multi-Agent AI System (Understanding, Knowledge, Response & Coaching, Quality Monitoring).
5. AI suggestions are shown to the agent in the AI Copilot Panel.

6. Agent sends a reply to the customer, informed by the suggestions.
7. Conversation is stored and analysed for ongoing context and history.
8. Reports and analytics are generated for the session and rolled up into cross-session trends.

4. Schema and Data Models

4.1 MongoDB Atlas — Collections

Users

Field	Type	Description
user_id	String (PK)	Unique identifier for the user.
name	String	Full name of the user.
email	String	Login Email, unique.
role	String	One of: support_agent, team_lead, quality_analyst, admin.
created_at	Timestamp	Account creation date and time.

Conversations

Field	Type	Description
conversation_id	String (PK)	Unique identifier for the conversation session.
agent_id	String (FK -> users)	Support agent handling the conversation.
mode	String	One of: simulator, manual, replay.
scenario	String	Product context and customer scenario description.
messages	Array	Ordered list of message objects (sender, text, timestamp, intent, sentiment).
status	String	One of: active, completed, escalated.
started_at/ended_at	Timestamp	Conversation start and end times.

Reports

Field	Type	Description
report_id	String (PK)	Unique identifier for the post-interaction report.
conversation_id	String (FK -> conversations)	Conversation this report summarizes.
sentiment_journey	Array	Sentiment value at each conversation turn.
quality_score	Number	Overall resolution quality score.

escalation_risk	String	Peak escalation risk level reached (low, medium, high).
coaching_recommendations	Array	Personalized coaching notes for the agent.
generated_at	Timestamp	Time the report was generated.

Feedback

Field	Type	Description
feedback_id	String (PK)	Unique identifier for the feedback entry.
conversation_id	String (FK -> conversations)	Conversation being reviewed.
reviewer_id	String (FK -> users)	Quality analyst or team lead giving feedback.
comments	String	Free-text review comments.
rating	Number	Manual quality rating given by the reviewer.

Analytics

Field	Type	Description
metric_id	String (PK)	Unique identifier for the analytics record.
scope	String	Aggregation level: agent, team, or organisation.
period	String	Reporting period (e.g. daily, weekly).
avg_quality_score	Number	Average resolution quality score for the scope and period.
escalation_rate	Number	Proportion of conversations that reached high escalation risk.
common_knowledge_gaps	Array	Frequently missing or low-confidence knowledge topics.

4.2 ChromaDB — Vector Collection

knowledge_base

Field	Type	Description
doc_id	String (PK)	Unique identifier for the source document or chunk.

embedding	Vector	Numerical embedding representation of the chunk text.
chunk_text	String	The underlying text chunk (FAQ, policy, or manual excerpt).
source	String	Origin document name (e.g. policy PDF, FAQ file).
metadata	Object	Additional tags: category, product, last_updated.

4.3 Firebase Authentication

Field	Type	Description
uid	String (PK)	Firestore-issued unique user identifier.
email	String	Authenticated email address.
role_claim	String	Custom claim mapping to application role.
last_login	Timestamp	Most recent successful sign-in.

5. Agent Specifications

The project defines six discrete agents. Four run-time agents form the live coaching pipeline; two additional agents support Simulator Mode and post-conversation reporting.

5.1 Customer Simulator Agent

Generates realistic customer messages turn by turn, based on a defined problem scenario and persona, with configurable emotional progression across the conversation.

5.2 Intent & Sentiment Analysis Agent

Identifies customer intent, emotional state, frustration level, and satisfaction trend on a per-message basis.

5.3 Knowledge Recommendation Agent (RAG Agent)

Performs RAG-powered retrieval of relevant support articles, FAQs, and troubleshooting steps using semantic search against the ChromaDB vector store.

5.4 Coaching & Response Suggestion Agent

Generates suggested agent responses, evaluates tone and clarity, and provides communication improvement tips on a per-turn basis.

5.5 Escalation Risk Monitor Agent

Continuously scores escalation likelihood with reasoning and triggers an alert overlay when risk crosses a high threshold, along with recommended intervention strategies.

5.6 Post-Interaction Summary Agent

Generates the interaction summary, sentiment journey timeline, overall resolution quality score, and personalised coaching recommendations after a conversation concludes.

6. Functional Modules

- **Session Configuration and Interaction Mode Selection Module.**
- **Support Knowledge Base Ingestion and RAG Pipeline Module.**
- **Customer Simulator Agent and Conversation Management Module.**
- **Multi-Agent Coaching Pipeline and Orchestration Layer.**
- **Live Support Console UI (Three-Panel Coaching Interface).**
- **Escalation Risk Detection and Alert Module.**
- **Post-Interaction Report and Performance Analytics Module.**

7. Project Phases

The build is planned across four two-week phases (milestones) spanning eight weeks in total.

Phase 1 — Weeks 1-2: Foundations and Knowledge Ingestion

- **Study customer support workflows, coaching methodologies, RAG architecture, and multi-agent design patterns.**
- **Design system architecture, agent responsibilities, orchestration flow, and data models.**
- **Build the session configuration module — interaction mode selection, product context, and customer scenario definition.**
- **Develop Support Knowledge Base ingestion — upload FAQs, support docs, and policy PDFs; implement chunking, embedding, and vector indexing via the RAG pipeline.**

Phase 2 — Weeks 3-4: Core Agents

- **Build the Customer Simulator Agent — realistic, scenario-consistent messages with configurable persona and emotional progression.**
- **Develop the Intent and Sentiment Analysis Agent — intent, emotional state, frustration level, and satisfaction trend per message.**
- **Build the Knowledge Recommendation Agent — contextually relevant retrieval from the**

RAG knowledge base per conversation turn.

- **Test Simulator Agent realism and Knowledge Agent retrieval accuracy across multiple support scenarios.**

Phase 3 — Weeks 5-6: Coaching, Escalation, and Console UI

- **Build the Coaching and Response Suggestion Agent** — suggested responses, tone and clarity evaluation, and communication tips per turn.
- **Develop the Escalation Risk Monitor Agent** — continuous scoring with reasoning, and a high-threshold alert overlay.
- **Build the Live Support Console UI** — three-panel layout with conversation window, real-time coaching feed, and knowledge panel.
- **Implement Manual Mode and Replay Mode.**

Phase 4 — Weeks 7-8: Reporting, Analytics, and Delivery

- **Build the Post-Interaction Summary Agent** — summary, sentiment journey timeline, resolution quality score, and coaching recommendations.
- **Develop the Performance Analytics Module** — cross-session trends including escalation triggers, knowledge gaps, and improvement indicators.
- **Conduct end-to-end testing across all three interaction modes, all agents, and post-interaction reporting.**
- **Prepare technical documentation, project report, and final demonstration.**

8. Evaluation Criteria

- **Realism and scenario consistency of Customer Simulator Agent generated messages.**
- **Accuracy of the Intent and Sentiment Analysis Agent across varied customer emotional states and intents.**
- **Relevance and retrieval quality of RAG-powered Knowledge Recommendation Agent suggestions.**
- **Usefulness and actionability of real-time coaching, response suggestions, and escalation alerts.**
- **Quality and clarity of post-interaction performance reports and sentiment journey visualisation.**

- **Completeness of implementation, testing, documentation, and final demonstration.**

9. Glossary

Term	Definition
RAG	Retrieval-Augmented Generation — retrieves relevant documents from a knowledge base and supplies them to an LLM as context before generation.
LLM	Large Language Model — the underlying AI model (e.g., GPT-4.1 / GPT-5) used for reasoning and text generation.
Vector Database	A database (ChromaDB) that stores content as numerical embeddings, enabling semantic similarity search.
Embedding	A numerical vector representation of text used to measure semantic similarity.
Escalation Risk	A continuously scored likelihood that a conversation will require intervention beyond the agent's standard resolution path.
Orchestration Layer	The coordination logic (via LangChain and the AI Orchestrator service) that sequences agent calls and aggregates their outputs.
Sentiment Journey	A timeline visualisation of the customer's sentiment across the course of a conversation.